



C.V. RAMAN GLOBAL UNIVERSITY
DEPARTMENT OF COMPUTER SCIENCE ENGINEERING

CITY CORPORATION AUTOMATION SYSTEM (CCAS)

Digitizing Municipal Services for a
Smarter, Cleaner and More Responsive City



CITIZEN
CENTRIC



TRANSPARENT
GOVERNANCE



FASTER
SERVICES



SMART
CITY MANAGEMENT



Course
Software Engineering Lab
(CS387)



Group
8



Academic Year
2026 – 27
Semester: 5th



Presentation
Case Study Report
City Corporation System



1. INTRODUCTION

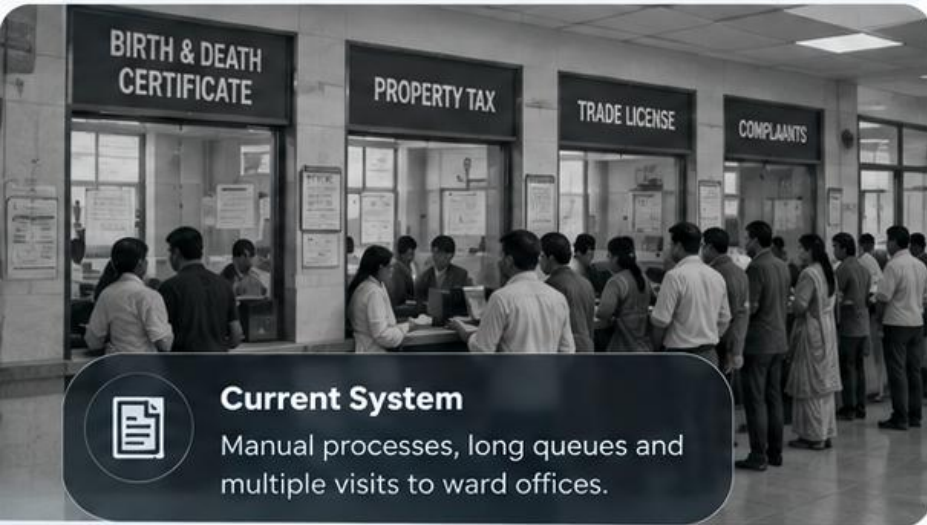
Introduction & Problem Statement

Understanding the need for a digital and citizen-centric city corporation system



Background

A city corporation is responsible for providing essential civic services such as issuing certificates and licences, collecting taxes and bills, maintaining infrastructure and resolving citizen grievances. In many cities, these services are still delivered through manual and paper-based processes.



Current System

Manual processes, long queues and multiple visits to ward offices.



Key Problems



Time & Effort

Citizens must visit the office physically, often multiple times.



Lack of Transparency

No easy way to track application or complaint status.



Data Duplication & Loss

Paper records may be misplaced or damaged.



Delayed Revenue Collection

Tax notices are sent late and payments are accepted only during office hours.



Weak Monitoring

Managers cannot easily view pending files or generate accurate reports.



Need for a Digital Solution

A centralized, web-based system can automate municipal services, reduce manual work, and provide citizens with a convenient, transparent and efficient way to access services.





2. PROPOSED SYSTEM

Proposed System & Objectives

A centralized, web-based platform to automate and streamline municipal services for citizens, employees, and administrators.

Proposed System (CCAS)

The City Corporation Automation System (CCAS) is a **centralized web and mobile based system** that digitizes and automates major municipal services. It connects citizens, municipal employees and administrators through a single platform, enabling online service applications, complaint management, payments, tracking and real-time notifications.



Key Objectives



Digitize Municipal Services

Automate key services such as certificates, licences, tax payments and complaints.



Convenient Citizen Access

Provide an easy-to-use online platform accessible from anywhere, anytime.



Reduce Manual Processing

Minimize paperwork and physical visits to offices.



Improve Transparency & Accountability

Enable real-time tracking and audit trails for all requests and actions.



Faster Service Delivery

Streamline workflows for quicker application and complaint resolution.

Smart Services for a Better City

Citizens | Services | Sustainability



Citizen Centric



Efficient Governance



Cleaner City



Data Driven Decisions



4. FEASIBILITY STUDY & PROCESS MODEL

Feasibility Study & Software Process Model

The system is feasible from technical, economic, operational and schedule perspectives.

Feasibility Study



Technical Feasibility

The required technology and infrastructure are readily available.

- ✓ Uses standard web technologies (HTML5, CSS3, JavaScript, Node.js)
- ✓ MySQL database and REST APIs
- ✓ Scalable and secure architecture
- ✓ Integration with payment, SMS and e-mail services is feasible



Economic Feasibility

The project is cost-effective and provides long-term benefits.

- ✓ Uses open-source technologies
- ✓ Reduces manual work and paperwork
- ✓ Lower operational costs
- ✓ Improved revenue collection
- ✓ High return on investment (ROI)



Operational Feasibility

The system is user-friendly and suitable for real-world use.

- ✓ Simple and intuitive interface
- ✓ Minimal training required
- ✓ Follows existing approval hierarchy in digital form
- ✓ Support and help section for users



Schedule Feasibility

The project is divided into increments and can be completed in a realistic timeframe.

- ✓ Implementation in three increments
- ✓ Each increment ~ 16 weeks
- ✓ Early delivery of core features
- ✓ Allows feedback and refinement

Incremental Process Model

The Incremental Process Model is used for CCAS, where the complete requirement set is understood at a high level and the system is built and delivered in a series of increments. Each increment goes through analysis, design, coding and testing, and adds new functionality to the product.

Increment 1

Core platform: login, roles, certificate & licence requests



Analysis



Design



Coding



Testing

Release 1

Increment 2

Property tax billing and online payment



Analysis



Design



Coding



Testing

Release 2

Increment 3

Complaint management, notifications and reports



Analysis



Design



Coding



Testing

Release 3



Why Incremental Model?



Early delivery of core services



User feedback in each increment



Risk reduction through phased delivery



Easier refinement of later features



5. REQUIREMENT ANALYSIS

Requirements & User Roles

Defining what the system should do and who will use it.

A unified platform
for citizens, employees
and administrators



User Roles and Responsibilities

Citizen

- ✓ Register and log in
- ✓ Apply for services (certificate, licence, etc.)
- ✓ Pay taxes and bills
- ✓ Submit and track complaints
- ✓ Track application status
- ✓ Receive SMS/e-mail notifications
- ✓ View and download certificates and receipts

Department Officer

- ✓ Verify submitted documents
- ✓ Approve or reject applications
- ✓ Assign complaints to field staff
- ✓ Update application and complaint status
- ✓ Communicate with citizens
- ✓ View department reports

Field Staff

- ✓ Receive assigned complaints
- ✓ Update progress status
- ✓ Upload photos as proof of resolution
- ✓ Communicate with officers and citizens
- ✓ Close complaints with remarks

Administrator

- ✓ Manage users and employees
- ✓ Manage services, departments and wards
- ✓ Monitor applications and complaints
- ✓ Generate reports and analytics
- ✓ Manage system settings
- ✓ View audit logs

Key Functional Requirements

 User Registration & Login FR-001 – FR-004	 Profile Management FR-005 – FR-007	 Municipal Service Management FR-008 – FR-013	 Complaint Management FR-014 – FR-020	 Property Tax Management FR-021 – FR-024	 Certificate Services FR-025 – FR-028	 Payment Management FR-029 – FR-032	 Notifications & Reports FR-033 – FR-035
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Key Non-Functional Requirements

 Performance Pages load < 3s Support 500+ concurrent users	 Security HTTPS, role-based access, input validation	 Usability Clean, responsive UI with multilingual support (future)	 Reliability & Maintainability 99% uptime, modular architecture, scalable system
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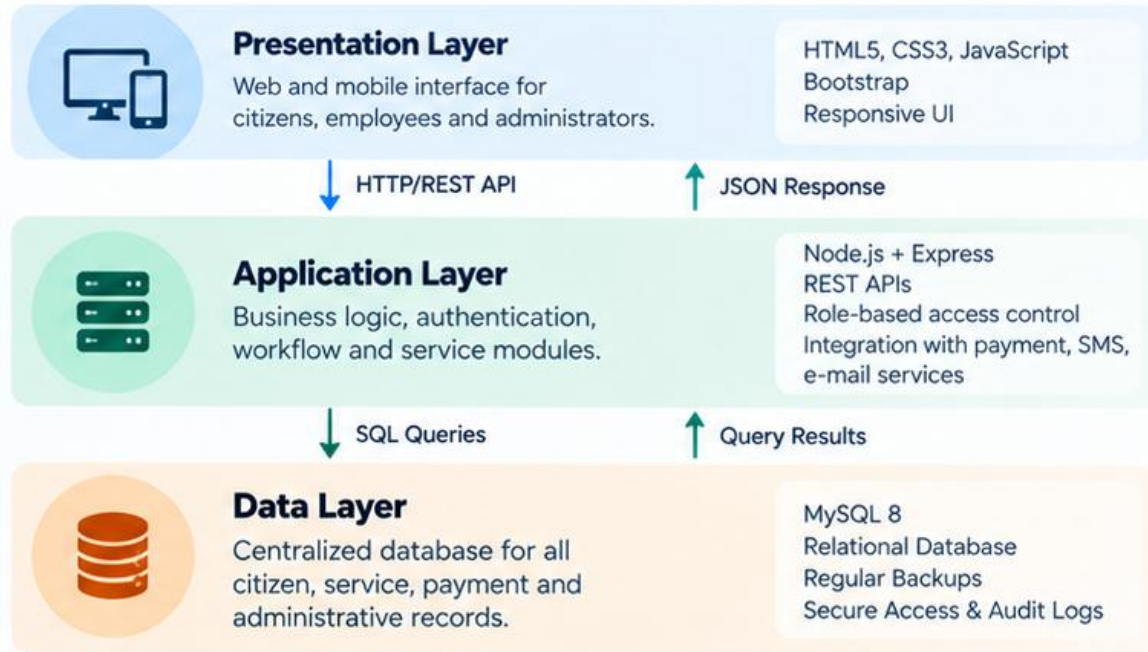


6. SYSTEM DESIGN

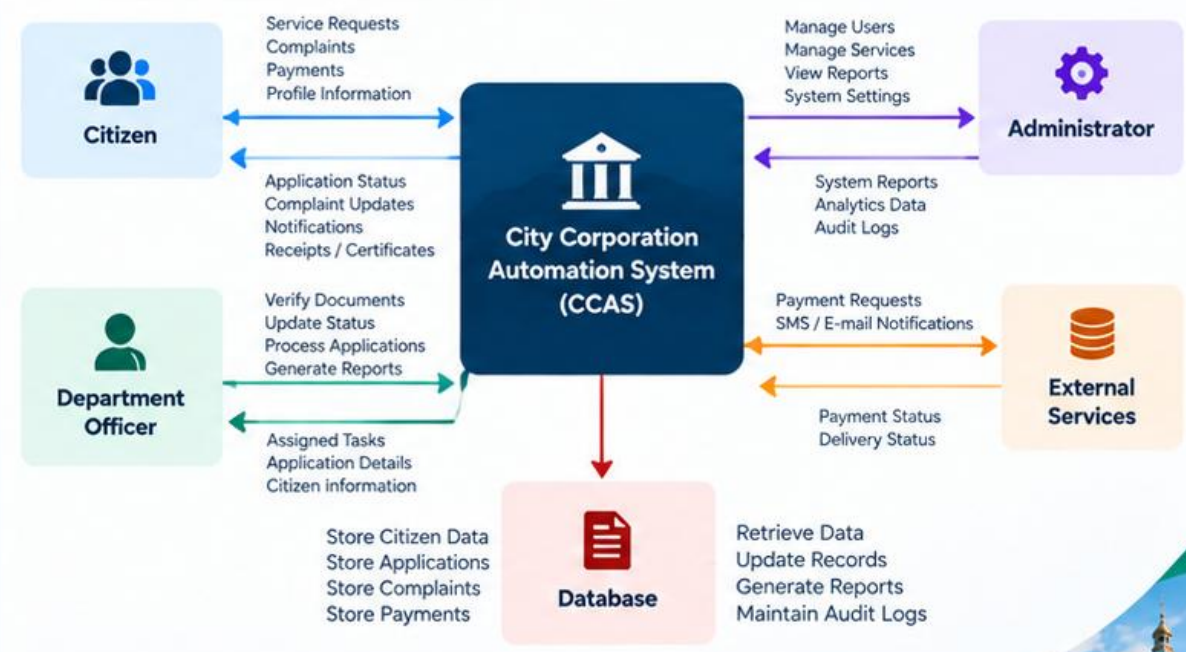
System Architecture & Data Flow

A secure, scalable and modular architecture to support municipal services.

System Architecture (Three-Layer Architecture)



Data Flow Diagram (Level 0)



Key Design Elements





7. MODULE DESCRIPTION

System Modules & Key Workflows

Integrated modules to deliver end-to-end municipal services for a smarter city.



Certificate & Licence Management

- ✓ Birth & death certificate
- ✓ Trade licence
- ✓ Building plan approval
- ✓ Document verification
- ✓ Downloadable digital certificates with QR code



Property Tax Management

- ✓ Property registration
- ✓ View outstanding dues
- ✓ Online tax payment
- ✓ Penalty calculation
- ✓ Digital receipts & payment history



Complaint Management

- ✓ Register complaints (roads, drainage, garbage, water, etc.)
- ✓ Track complaint status
- ✓ Assign to department/field staff
- ✓ Update with remarks and photos
- ✓ Citizen notifications



Water & Sanitation Services

- ✓ Apply for new connection
- ✓ View and pay water bills
- ✓ Report water supply issues
- ✓ Schedule maintenance
- ✓ Track service requests



User & Employee Management

- ✓ Citizen registration & profile management
- ✓ Employee role management
- ✓ Department and ward management
- ✓ Role-based access control
- ✓ Audit logs



Reports & Analytics

- ✓ Service-wise reports
- ✓ Revenue collection reports
- ✓ Complaint resolution analytics
- ✓ Department performance
- ✓ Audit and activity logs

Complaint Management Workflow (Example)



1. Complaint Registration

Citizen submits complaint with description and photos.*



2. Assignment

System assigns to the concerned department/field staff.



3. Action & Update

Field staff updates progress, uploads proof and adds remarks.



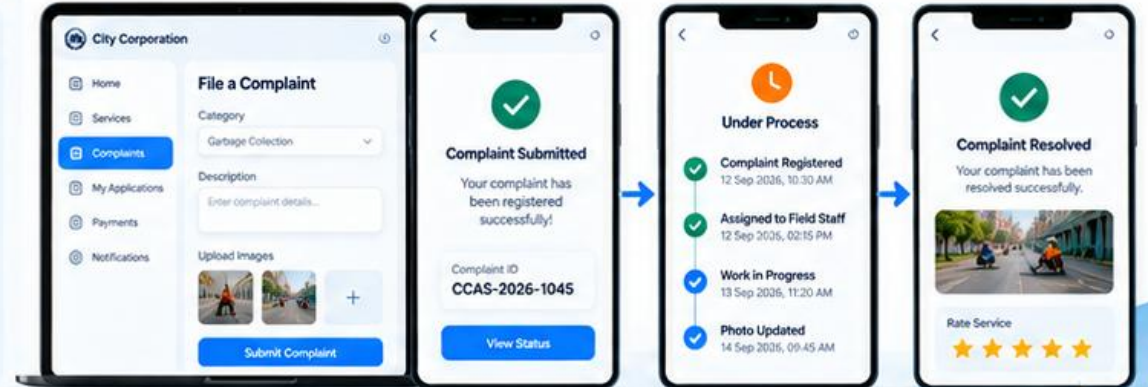
4. Resolution

Complaint marked as resolved with details and citizen feedback.



5. Notification

Citizen receives SMS/e-mail notification at every stage.



Module Overview



6+
Core Modules



30+
Key Features



3
User Types



End-to-End
Service Digitalization



Faster
Service Delivery





8. TECHNOLOGY STACK

Technology Stack & Infrastructure

Modern, scalable and secure technologies to build a reliable municipal platform.



Frontend

Responsive and user-friendly web and mobile interface



HTML5



CSS3



JavaScript



Bootstrap



React.js



Backend

Robust and scalable server-side application



Node.js



Express.js



Database

Secure and reliable data storage



MySQL 8



Hosting & Deployment

Scalable and highly available infrastructure



AWS



Docker



Tools & Others

Development, testing and project management



GitHub



Postman



VS Code



Figma



Jira



Nginx

System Infrastructure



Client Layer

Web Browser /
Mobile App
(Citizens, Staff,
Administrators)

HTTPS



Web Server

Nginx / Apache
(Reverse Proxy)

Requests
(REST API)



Application Server

Node.js + Express.js
(Business Logic
& APIs)

SQL
Queries



Database Server

MySQL 8
(Structured Data)

External Services



Payment Gateway
(Online Payments)



SMS / Email Service
(Notifications)



Cloud Storage
(Documents & Pictures)

Key Benefits



Secure & Reliable

Follows industry best practices
and secure architecture.



Scalable

Can handle increasing users
and services.



Flexible

Modular architecture for easy
maintenance and future upgrades.



Cost-Effective

Uses open-source technologies
and cloud infrastructure.

Technology Highlights



Modern
Web Technologies



Secure
& Scalable



Open Source
& Cost Efficient



Cloud Ready



Easy Integration
with External Services



9. TESTING & IMPLEMENTATION

Testing Strategy & Implementation Plan

Ensuring a reliable, secure and successful deployment for real-world use.

Testing Strategy



Unit Testing

Test individual modules and functions.



Integration Testing

Test interaction between modules.



System Testing

Test the complete system end-to-end.



User Acceptance Testing (UAT)

Testing by actual end users (citizens and municipal staff).



Security Testing

Check for vulnerabilities and unauthorized access.



Performance Testing

Test system under high load and multiple users.

Testing Environment



Test Case Summary

Test Type	No. of Test Cases	Pass	Fail	Status
Unit Testing	25	24	1	Passed
Integration Testing	18	18	0	Passed
System Testing	20	19	1	Passed
User Acceptance Testing	15	15	0	Passed
Security Testing	10	10	0	Passed
Performance Testing	8	7	1	In Review

Implementation Plan



Success Metrics



Ready for a
Smarter City



Reliable
System



Citizen-Centric
Services



Scalable
Deployment



Measurable
Impact



10. CONCLUSION

Conclusion Open Source is the Best Choice

A reliable, scalable, secure and cost-effective solution for modern municipal systems.



Cost-Effective

No licensing fees, reducing long-term costs and enabling better resource utilization.



Flexible & Customizable

Can be tailored to meet specific municipal requirements and adapted for future needs.



Secure & Reliable

Transparent code, active community support and regular updates ensure a robust and secure system.



Scalable

Easily accommodates growing users, data and new services without high additional costs.



Community Support

Backed by a large global community, continuous improvements and wide availability of resources.



Open Source Technologies Outperform Proprietary Solutions

Considering cost, flexibility, security, scalability and long-term sustainability, open source technologies are the best choice for building and maintaining the City Corporation Automation System (CCAS). They provide a powerful, future-ready and community-driven ecosystem, making them superior to proprietary (paid) solutions for municipal applications.

“Open source is not just an alternative, it is **the better choice** for a smarter, more transparent and sustainable municipal future.”



Lower Cost
Higher Value



Freedom
to Innovate



Secure
and Transparent



Scales with
Future Needs



Powered by
a Global Community



Open Source Today
for a Smarter Tomorrow